



What's Cognitive Manufacturing? Why Should It Matter To You?

Due to unavailability of time-critical information, challenges like operational efficiency and safety affect the productivity of employees and the overall industry of manufacturing. With its ability to organise massive amounts of data, AI based cognitive manufacturing systems are highly valuable. But how far would they solve problems? Would cognitive manufacturing be truly beneficial?

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Big data and artificial intelligence (AI) are giving a huge boost to Industry 4.0. Intelligent software solutions based on AI models can process high volumes of data generated by an industry to identify trends and patterns that can be used to make manufacturing processes more efficient and reduce their energy consumption. More factories are instrumented with IoT, resulting in an increase of the data volume.

To process this large influx of data and deal with the complexity of the analytics,

conventional computing struggles. To pave the way forward to Industry 4.0 and beyond, cognitive manufacturing has to evolve.

Cognitive manufacturing infrastructure is built on the foundation of IoT and analytics. It uses the data across systems, equipment, and processes to derive actionable insight across the entire value chain, starting from design through manufacture till support.

First, let's have a look at the challenges that current manufacturing processes face, and how these challenges affect the profit for industries.

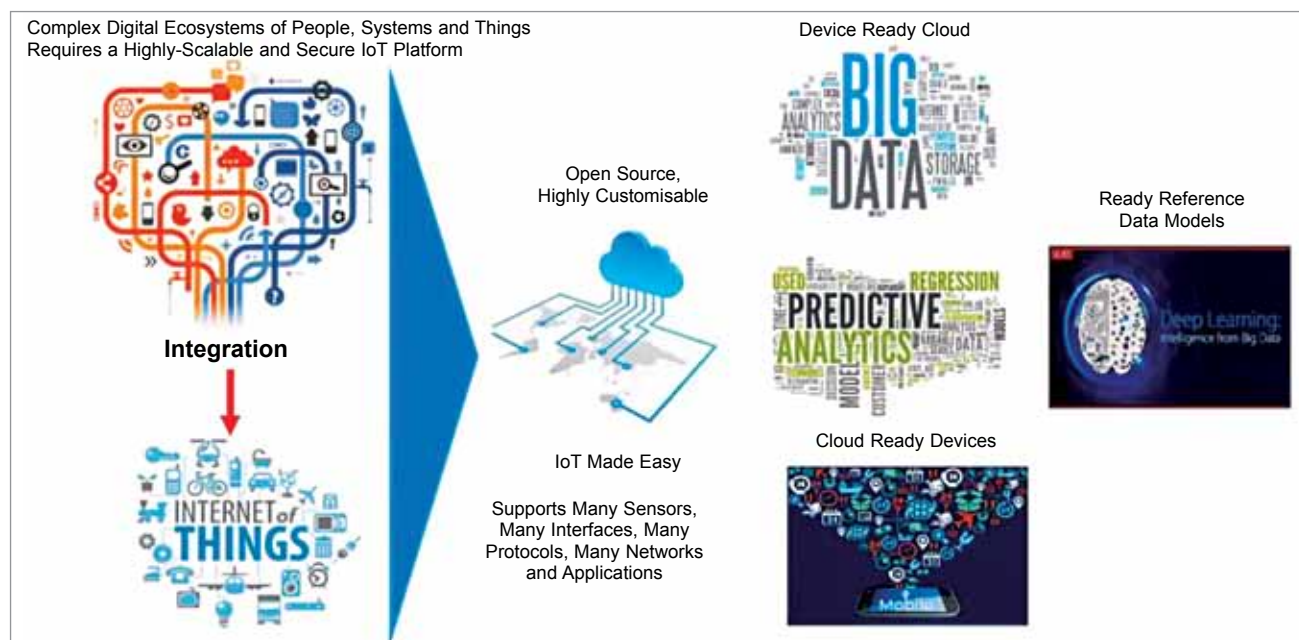


Fig. 1: A pictorial representation of the challenges faced by Industry 4.0 (Courtesy: LivNSense)